Design and Analysis of Absorptive Reflectionless Band Pass Filters

A Report

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by

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I would further like to acknowledge the value of the published literature that guided this work. The study of reflectionless and absorptive filters by Lemire and the earlier conceptual foundations provided by Morgan and Boyd clarified how passive networks can be designed to absorb rather than reflect rejected energy. Those ideas formed the theoretical basis for this thesis.

Abstract

Absorptive reflectionless filters are passive RF and microwave networks designed to control not only spectral selectivity but also the fate of rejected energy. In conventional band pass filters, stopband signals are typically reflected back toward the source, which can create mismatch, instability, and unwanted coupling between cascaded stages. Reflectionless filters address this limitation by routing rejected energy into internal resistive paths so that the network remains well matched across a much wider frequency range. This property is particularly valuable in receiver front ends, instrumentation chains, and tightly coupled RF subsystems where stopband reflections can degrade system-level performance.

This thesis presents the design and analysis of absorptive reflectionless band pass filters using lumped-element synthesis techniques. The work combines even-odd mode analysis, symmetry arguments, and Cauer ladder duality to derive a practical filter methodology. The resulting designs are examined through a single-stage baseline cell, a cascaded multi-stage topology, and a comparative response overlay. The S-parameter responses are interpreted using S_{11} and S_{21} to evaluate both the passband transmission and the broadband matching behavior.

The report demonstrates that reflectionless band pass filters can provide useful selectivity while maintaining low reflection in both the passband and stopband. Cascading improves skirt steepness and stopband rejection, while scaling confirms that the topology is transferable across different frequency ranges. The overall conclusion is that reflectionless band pass filters offer a rigorous and system-oriented alternative to conventional reflective filtering approaches in RF design.

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Nomenclature

S_{11}	Input reflection coefficient.
S_{21}	Forward transmission coefficient.
S_{12}	Reverse transmission coefficient.
S_{22}	Output reflection coefficient.
Z_0	Reference impedance, typically $50\,\Omega$.
Z_{in}	Input impedance of the network.
Γ	Reflection coefficient at the input port.
ω_0	Center angular frequency.
B	Bandwidth parameter in band-pass transformation.
Q	Quality factor of a resonant network.
L	Inductance.
C	Capacitance.
R	Resistance.

Chapter 1

Introduction

1.1 Background

Filters are fundamental building blocks in RF and microwave systems because they determine which spectral components are permitted to pass and which are suppressed. In communication receivers, transmitters, radar front ends, and instrumentation systems, the filter is often located at a critical interface between the external environment and a sensitive active circuit. Its performance influences not only signal integrity but also impedance matching, dynamic range, and system stability.

Conventional band pass filters are usually optimized to maximize transmission in the desired band and attenuate out-of-band signals. This objective is appropriate when the only concern is spectral selectivity. However, in practical systems, rejected energy does not disappear; it must either be reflected, dissipated, or transmitted elsewhere. If the filter reflects a large amount of stopband energy, the source sees a mismatch that can disturb the behavior of previous stages. The issue becomes more serious when filters are cascaded, because a reflection from one stage can interact with another and create unintended coupling paths.

Absorptive reflectionless filters address this limitation by changing the stopping mechanism. Instead of allowing rejected energy to be reflected back toward the source, the topology routes it into internal resistive branches. The network therefore remains broadband matched while still providing frequency selectivity. This dual behavior makes reflectionless filters especially useful in systems where both low reflection and meaningful attenuation are required.

1.2 Problem Statement

The central engineering problem in this thesis is the design of a band pass filter that provides selective transmission in the desired band while maintaining a low reflection coefficient across a much wider frequency span than conventional designs. Traditional filters can satisfy the passband requirement, but their stopband match is often poor. As a result, they reflect out-of-band energy rather than absorb it, which can degrade performance in RF signal chains.

The task is therefore not merely to construct a filter with a passband and a stopband, but to construct a filter whose impedance behavior is controlled in both regions. The desired network should transmit useful energy efficiently, absorb unwanted energy internally, and remain compatible with cascaded operation. Achieving this requires a synthesis approach grounded in symmetry, even-odd mode analysis, and a suitable prototype transformation.

1.3 Objectives

The objectives of this thesis are:

1. To study the theoretical basis of absorptive and reflectionless filtering.
2. To derive the band pass response from symmetric passive network concepts.
3. To evaluate the reflectionless behavior using S-parameters and impedance matching conditions.
4. To develop a single-stage reflectionless band pass filter as a reference implementation.
5. To extend the design into cascaded and frequency-scaled variants.
6. To compare the resulting responses and identify the key design trade-offs.

1.4 Scope

The scope of this thesis is limited to lumped-element reflectionless band pass filters intended for RF and microwave applications. The report focuses on design theory, mathematical formulation, circuit-level interpretation, and S-parameter-based analysis. It does not attempt a full electromagnetic packaging study or PCB manufacturing qualification.

Instead, the emphasis is on a rigorous synthesis path and on the interpretation of the resulting responses.

1.5 Thesis Organization

Chapter 2 reviews conventional filters and the evolution toward absorptive topologies. Chapter 3 develops the theoretical foundations, including S-parameters, even-odd mode analysis, and ladder duality. Chapter 4 describes the synthesis and methodology used to derive the practical filter variants. Chapter 5 discusses circuit implementation and component choices. Chapter 6 presents the simulated and experimental responses. Chapter 7 discusses design trade-offs and practical implications. Chapter 8 concludes the thesis and outlines future work.

Chapter 2

Literature Review

2.1 Conventional Band-Pass Filter Synthesis

Classical band pass filters are commonly synthesized from normalized low-pass prototypes and then transformed to the desired frequency band. The standard design families are Butterworth, Chebyshev, and elliptic responses. Each offers a distinct compromise between passband flatness, transition steepness, and implementation complexity. Butterworth filters are maximally flat, Chebyshev filters introduce controlled ripple to obtain a sharper transition, and elliptic filters achieve the steepest selectivity by introducing transmission zeros.

These classical responses are highly effective when the design objective is forward transmission. However, they do not directly address the stopband impedance presented to the source. A conventional band pass filter may provide excellent attenuation and still reflect substantial energy outside the passband. The classical synthesis objective is therefore incomplete when the system also requires broadband matching.

2.2 Return Loss and Stopband Mismatch

The mismatch behavior of a filter is captured through the reflection coefficient. For a two-port network, the input return loss is defined as

$$RL(f) = -20\log_{10}|S_{11}(f)|\,. \tag{2.1}$$

If the network is well matched, then S_{11} is small and the return loss is large in magnitude. In conventional band pass filters, this condition is often met only within the passband. Outside the passband, the input impedance becomes reactive and the filter can behave as a poor match, which increases reflection.

This behavior is not only a measurement issue; it is a system issue. Reflected energy can travel backward through the signal chain, interact with source impedance, and disturb the behavior of previous components. In multi-stage RF systems, the mismatch of one block can influence the operating conditions of another block. As a result, the design objective must expand beyond selectivity to include the management of rejected energy.

2.3 Evolution Toward Absorptive and Reflectionless Networks

Reflectionless and absorptive filters were introduced to address the mismatch problem by embedding resistive dissipation paths into the topology. The key idea is that out-of-band energy should be absorbed internally rather than returned to the source. Lemire demonstrated a systematic realization of this idea in a symmetric passive network and showed that the resulting filter can be implemented with strong agreement between simulation and measurement [1].

Morgan and Boyd provided the conceptual framework for this development by showing how symmetry, duality, and complementary half-circuit behavior can be used to construct passive networks that are absorptive in the stopband [2]. Their contribution is important because it does not merely describe a desirable response; it explains how the topology can be synthesized. This shift from descriptive to constructive design is the foundation of the reflectionless filter approach used in this thesis.

2.4 Symmetry and Cascadability

A major advantage of reflectionless topologies is their symmetry. If each section is designed to remain matched, then multiple sections can be cascaded with less risk of mutual loading or inter-stage reflection. This property is particularly useful in RF filtering, where practical systems often require more than one stage of selectivity.

The modular nature of the topology also makes frequency scaling more straightforward. If the synthesis logic is based on a repeatable symmetric cell, then the same cell can often be reused at different center frequencies by adjusting the resonant elements. This flexibility is a major reason why reflectionless networks remain an active and useful topic in modern filter design.

The local workspace documents also reinforce the same implementation trend, especially the cascaded prototypes and comparative RF plots that appear in the internship files [3, 4, 5, 6, 7].

2.5 Summary of Literature Gaps

The literature establishes the existence and practical value of absorptive filters, but it still leaves room for a thesis-level synthesis and interpretation of the design flow. In particular, a structured presentation is needed that connects conventional filter theory, reflectionless topology, practical scaling, and the S-parameter behavior of single-stage and cascaded realizations. This thesis addresses that need by organizing the topic into a theory-driven design workflow.

Chapter 3

Theoretical Foundations

3.1 Two-Port Network Description

A passive RF filter is naturally modeled as a two-port network. The behavior of the network is often described through scattering parameters because these are directly measurable with a vector network analyzer and they relate naturally to reflection and transmission.

For a two-port network, the S-parameters are defined as

$$S_{11} = \left.\frac{b_1}{a_1}\right|_{a_2=0}, \qquad S_{21} = \left.\frac{b_2}{a_1}\right|_{a_2=0}, \tag{3.1}$$

$$S_{12} = \left.\frac{b_1}{a_2}\right|_{a_1=0}, \qquad S_{22} = \left.\frac{b_2}{a_2}\right|_{a_1=0}. \tag{3.2}$$

Here, a_1 and a_2 are incident waves and b_1 and b_2 are reflected waves. The input reflection coefficient is directly linked to S_{11}, while the forward transmission coefficient is represented by S_{21}.

3.2 Reflection Coefficient and Power Balance

The reflection coefficient at the input port is

$$\Gamma = \frac{Z_{\text{in}} - Z_0}{Z_{\text{in}} + Z_0}. \tag{3.3}$$

For a matched network, $\Gamma = 0$, which implies $Z_{\text{in}} = Z_0$. In a reflectionless filter, the goal is to maintain this condition over a broad frequency range rather than only inside the passband.

The energy balance of a passive network can be written as

$$P_{\text{in}} = P_{\text{load}} + P_{\text{loss}} + P_{\text{ref}}. \tag{3.4}$$

For a well-designed reflectionless filter, the reflected component P_{ref} is intentionally minimized while the unwanted stopband energy is absorbed in P_{loss}. This absorption is not a side effect; it is a design objective.

3.3 Even-Odd Mode Analysis

The reflectionless topology is symmetrical and therefore amenable to even-odd mode analysis. For a symmetric two-port, the port voltages and currents can be decomposed as

$$V_e = \frac{V_1 + V_2}{2}, \qquad V_o = \frac{V_1 - V_2}{2}, \qquad I_e = \frac{I_1 + I_2}{2}, \qquad I_o = \frac{I_1 - I_2}{2}. \tag{3.5}$$

The even-mode excitation corresponds to equal-phase driving at the ports, while the odd-mode excitation corresponds to opposite-phase driving. This decomposition reduces the original two-port problem to two simpler half-circuit problems.

In reflectionless filter synthesis, the even-mode path is typically associated with the desired transmission behavior, whereas the odd-mode path contributes the complementary absorptive behavior. The symmetry of the network ensures that these two responses combine in a way that preserves matching while creating the desired band-pass transmission.

3.4 Cauer Ladder Networks and Duality

Classical ladder prototypes are an important foundation for reflectionless filter synthesis because they provide a compact passive realization with alternating reactive elements. The low-pass prototype can be transformed to a band-pass form using the standard mapping

$$s \to \frac{s^2 + \omega_0^2}{Bs}, \tag{3.6}$$

where ω_0 is the center angular frequency and B is the bandwidth parameter.

Duality is then used to reinterpret the ladder so that the stopband energy can be directed into resistive branches. The key insight is that the network is not designed only for transmission, but for the simultaneous management of transmission and absorption. This is what distinguishes a reflectionless ladder-derived filter from a conventional one.

3.5 Frequency Scaling

If a prototype is synthesized for one center frequency, it may be scaled to another by using standard frequency and impedance transformation rules. For frequency scaling factor K_f and impedance scaling factor K_z, the element values transform as

$$L' = \frac{K_z}{K_f} L, \qquad C' = \frac{1}{K_z K_f} C. \tag{3.7}$$

This scaling principle is central to frequency adaptation in reflectionless networks. The same topology can therefore be reused at a different operating band while preserving the essential response shape.

3.6 Theoretical Summary

The theory of reflectionless filters combines several classical RF design ideas: symmetry, even-odd decomposition, ladder synthesis, and scaling. The novelty lies in the absorptive objective. Instead of allowing rejected energy to be reflected, the filter is designed so that stopband energy is terminated internally. This gives the network a more stable interface behavior and makes it especially suitable for cascaded RF systems.

Chapter 4

Methodology and Filter Design

4.1 Design Philosophy

The design philosophy adopted in this thesis is to begin with a simple matched cell and then extend it into more complex variants without breaking the underlying symmetry. The target is not only a useful band-pass response, but a response that remains well matched in the stopband. The methodology therefore treats transmission and absorption as coequal design goals.

4.2 Design Equations and Specification Summary

For a band-pass response with lower and upper cut-off frequencies f_{c1} and f_{c2}, the center frequency, bandwidth, and fractional bandwidth are given by

$$f_0 = \sqrt{f_{c1} f_{c2}}, \qquad BW = f_{c2} - f_{c1}, \qquad FBW = \frac{BW}{f_0}. \tag{4.1}$$

If the transmission-zero pair is located at f_{tz1} and f_{tz2}, then the transmission-zero bandwidth is

$$BW_{tz} = f_{tz2} - f_{tz1}. \tag{4.2}$$

These quantities are used to state the target passband and the stopband rejection window in a compact engineering form.

For a normalized low-pass prototype transformed into a band-pass section, the element values of a resonant branch can be written as

$$L_s = \frac{R_0 g_k}{\omega_0 FBW}, \qquad C_s = \frac{FBW}{R_0 \omega_0 g_k}, \tag{4.3}$$

for a series resonant branch, and

$$L_p = \frac{R_0 FBW}{\omega_0 g_k}, \qquad C_p = \frac{g_k}{R_0 \omega_0 FBW}, \tag{4.4}$$

for a shunt resonant branch. Here, R_0 is the system reference impedance and g_k is the normalized prototype coefficient. These relations provide the starting point for selecting inductor and capacitor values in the reflectionless band-pass synthesis.

The transmission-zero span is also useful when assessing stopband suppression, and it is given by

$$BW_{tz} = f_{tz2} - f_{tz1} = 441\,\text{MHz}. \tag{4.5}$$

For the implemented circuit shown in this report, the extracted element values are $L_1 = 18\,\text{nH}$, $L_2 = 15\,\text{nH}$, $C_1 = 6\,\text{pF}$, and $C_2 = 7.2\,\text{pF}$. Using the synthesized response, the lower and upper cut-off frequencies, transmission-zero band, and bandwidth are summarized in Table 4.1.

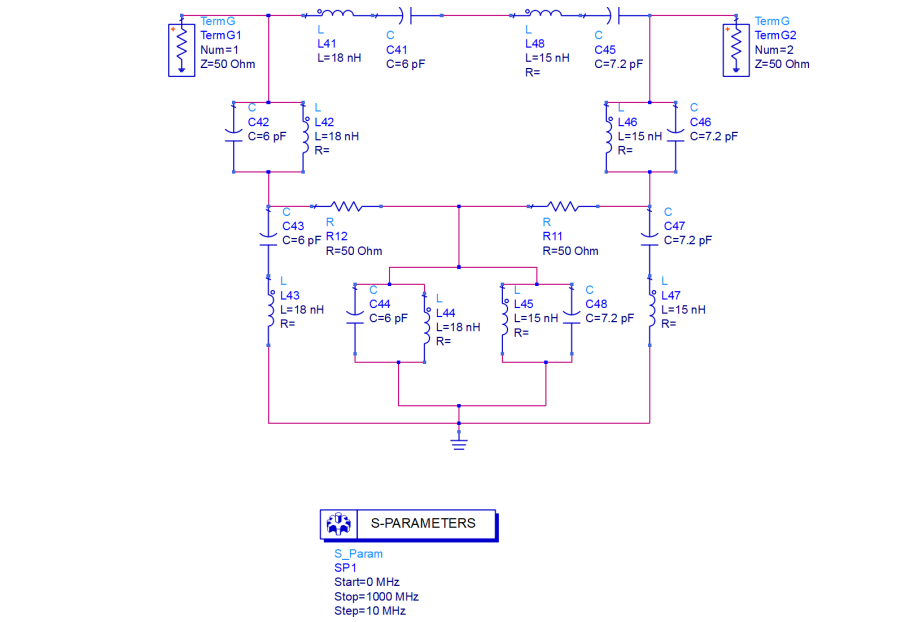
Table 4.1: Extracted reflectionless band-pass specifications for the implemented circuit.

Filter Number	**Cut-off Frequencies** (f_{c1}–f_{c2})	**Transmission Zero Frequencies** (f_{tz1}–f_{tz2})	**Cut-off BW** ($f_{c2} - f_{c1}$)	**TZ BW** ($f_{tz2} - f_{tz1}$)	**Element Values**
BPF-2	376–624 MHz	312–753 MHz	248 MHz	441 MHz	$L_1 = 18$ nH, $L_2 = 15$ nH, $C_1 = 6$ pF, $C_2 = 7.2$ pF

4.3 Single-Stage Reflectionless Cell

The first design step is a single-stage reflectionless cell. It serves as the baseline implementation from which other variants are derived. The cell contains resonant branches that define the band-pass transfer behavior and resistive paths that absorb rejected energy. The symmetric arrangement is essential, because it enables even-odd mode analysis and preserves the matched behavior of the network.

As illustrated in Fig. 4.1, the schematic and response form the reference design used throughout the thesis.



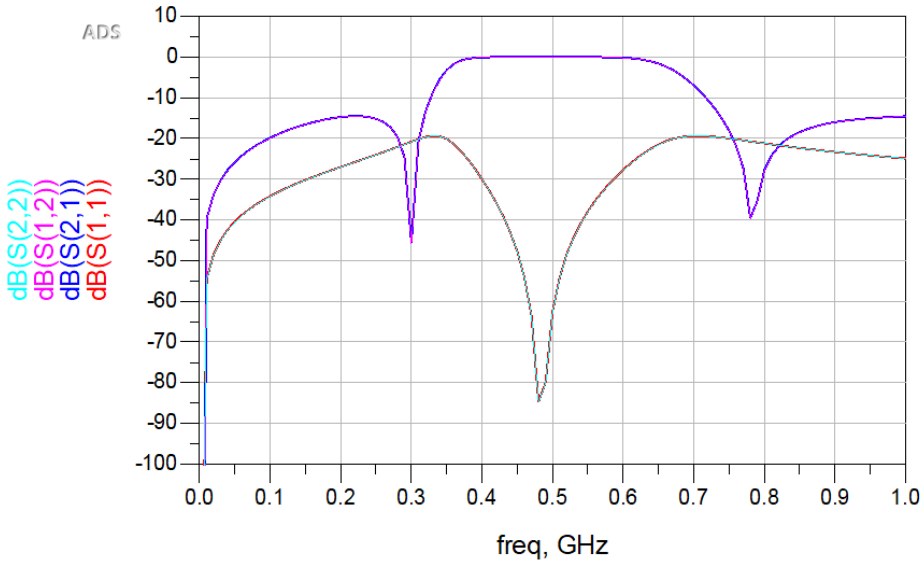


Figure 4.1: Baseline reflectionless band-pass cell used as the reference implementation. As illustrated in Fig. 4.1 and Fig. 4.1, the network establishes the core absorptive behavior that is preserved in later scaled and cascaded forms.

The upper panel shows the single-stage schematic, and the lower panel shows the corresponding S-parameter response.

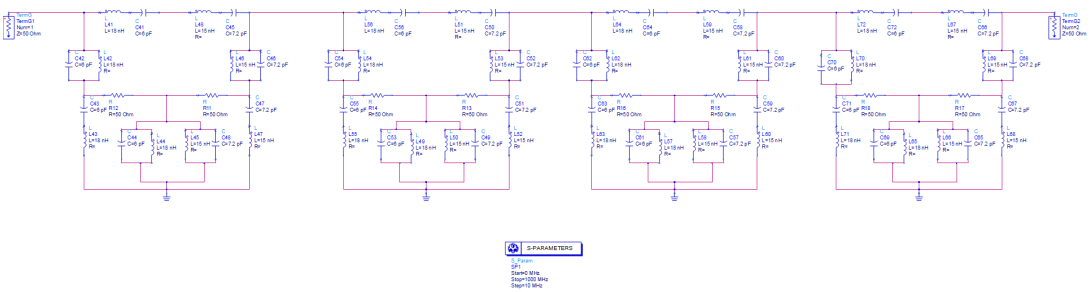
4.4 Cascaded Multi-Stage Realization

A key advantage of reflectionless cells is that they can be cascaded without creating strong inter-stage mismatch. If the single-stage transfer function is $H_1(s)$, then the idealized N-stage cascade can be expressed as

$$H_N(s) = \prod_{k=1}^{N} H_k(s), \qquad \text{and for identical stages, } H_N(s) = [H_1(s)]^N . \tag{4.6}$$

This formulation shows why the topology is attractive for higher-order filtering: the response can be sharpened by repetition while preserving the matched nature of each stage.

The cascaded schematic and response are shown in Fig. 4.2.



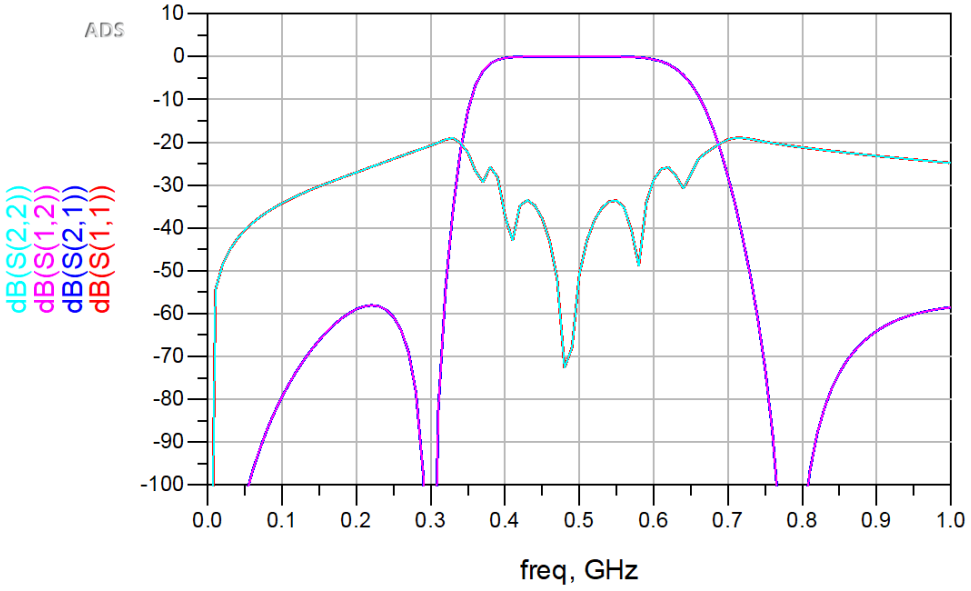


Figure 4.2: Cascaded multi-stage implementation. As illustrated in Fig. 4.2 and Fig. 4.2, the repeated reflectionless cell improves skirt sharpness while keeping the broadband match intact.

The upper panel shows the cascaded schematic, and the lower panel shows the corresponding response.

4.5 Design Summary

The methodology can be summarized as a sequence of linked steps: start from a symmetric matched cell, analyze it using even-odd mode decomposition, implement the desired band-pass function using ladder duality, and then extend the response through cascading. This framework is systematic and extensible, which makes it suitable for both theoretical study and practical implementation.

Chapter 5

Circuit Implementation

5.1 Component Selection

A practical reflectionless filter must be realized using components that support the intended resonant behavior while keeping parasitic effects manageable. In a lumped-element implementation, inductors should exhibit sufficiently high self-resonant frequency and low series resistance, while capacitors should have low equivalent series inductance and stable dielectric characteristics. Resistive elements must be chosen carefully because they directly influence the absorptive path and the overall insertion loss.

In the context of reflectionless filters, the component count and the number of unique values are also important. One advantage of the symmetry-based synthesis is that the network can often be realized with a small number of repeated values, which simplifies procurement and improves repeatability.

5.2 Layout Considerations

At RF frequencies, the layout is part of the circuit. Even when the topology is correct in schematic form, unintended coupling, stray inductance, pad capacitance, and trace discontinuities can significantly alter the response. For this reason, short interconnects, careful grounding, and compact placement of resonant components are essential.

The absorptive nature of the filter makes layout especially important because any unintended mismatch can interfere with the broadband matching objective. If the physical realization introduces parasitic resonances, the stopband return loss may deteriorate even if the ideal circuit is well behaved.

5.3 Single-Stage Physical Realization

The single-stage layout is the simplest implementation and is therefore the most practical starting point. It serves as the physical reference for the design process and is the easiest topology to verify in simulation. The corresponding schematic in Fig. 4.1 provides the basis for the later cascaded and scaled forms.

5.4 Cascaded Implementation Strategy

The cascaded structure is obtained by repeating the same matched cell multiple times. Because the stage interfaces are intended to be matched, each additional section contributes selectivity without forcing a large interstage reflection penalty. This is one of the most important practical strengths of the reflectionless approach.

5.5 Implementation Summary

The implementation stage confirms that the synthesis approach is modular. A single matched cell can be used as a building block, and more complex responses can be constructed by repetition or scaling. This modularity is one of the main reasons reflectionless filters are attractive for RF design practice.

Chapter 6

Simulation and Experimental Results

6.1 Evaluation Metrics

The principal metrics used to evaluate the filter responses are insertion loss, return loss, stopband attenuation, and the overall shape of the S_{21} and S_{11} curves. A useful reflectionless filter should transmit the desired band efficiently while maintaining a low reflection coefficient outside the passband.

The quality of the design is therefore assessed not only by the depth of attenuation, but by the behavior of the input match across the frequency span. This is what distinguishes absorptive topologies from conventional reflective filters.

6.2 Single-Stage Response

The single-stage response establishes the baseline behavior of the filter family. It shows the intended passband transmission along with broadband matching. Figure 4.1 demonstrates the characteristic relationship between the transmission and reflection curves: the passband is preserved while the stopband energy is not strongly reflected.

6.3 Cascaded Response

The cascaded response highlights the main trade-off in reflectionless design: sharper selectivity is achieved when multiple sections are combined, but the cumulative insertion loss can increase. The benefit is a more selective skirt and stronger stopband suppression, which is shown in Fig. 4.2.

6.4 Comparative Analysis

The comparative plot in Fig. 6.1 overlays the single-stage and cascaded responses. It makes the primary trade-off clear: the cascaded implementation improves skirt steepness and stopband attenuation, while the single-stage implementation remains simpler and lower in complexity.

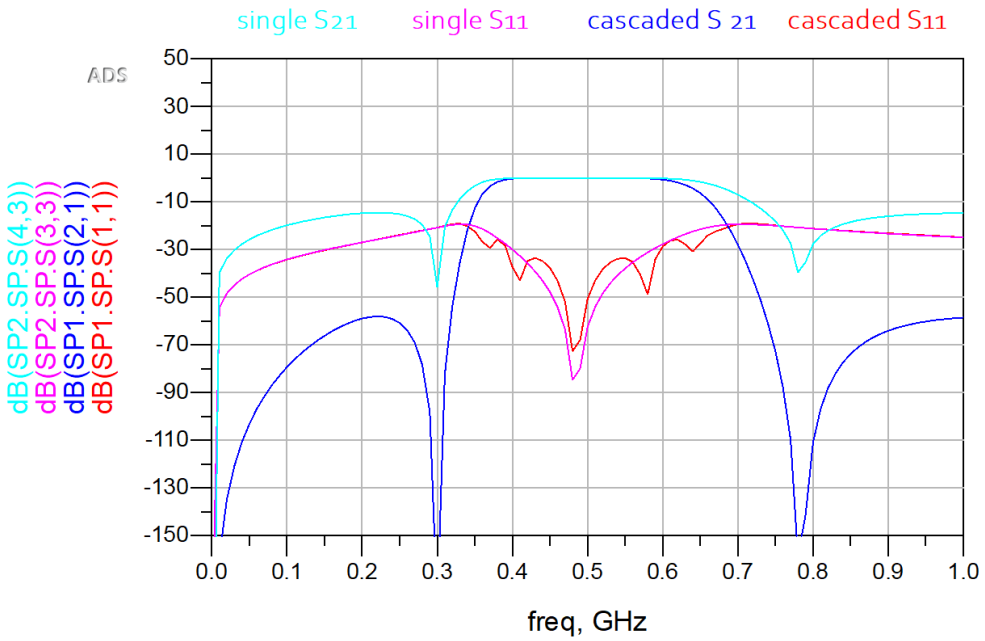


Figure 6.1: Overlay comparison of the single-stage and cascaded designs. The figure highlights the stronger skirt attenuation of the cascaded implementation and the broadband match preserved by both topologies.

Table 6.1: Qualitative comparison of the reflectionless design variants.

Metric	Single stage	Cascaded
Passband selectivity	Moderate	High
Stopband matching	Good	Very good
Implementation effort	Low	Higher
Best use case	Baseline cell	Higher rejection

6.5 Discussion of Results

The results support the main design claim of this thesis: rejected energy can be absorbed instead of reflected when the topology is constructed appropriately. This is valuable in RF systems because it reduces the likelihood that out-of-band energy will propagate backward

into previous stages. The matched behavior of the reflectionless filter is therefore not only a desirable electrical characteristic but also a system-level advantage.

Chapter 7

Discussion

7.1 Trade-Offs

The primary trade-off in reflectionless band pass design is between selectivity and loss. A cascaded structure can improve attenuation and sharpen the transition bands, but it may also increase insertion loss and physical complexity. A single-stage design is simpler and easier to realize, but it may not provide the same degree of stopband rejection.

Another important trade-off is between ideal synthesis and physical realization. Practical components introduce parasitics, tolerance spread, and frequency-dependent losses. These nonidealities can shift the response and degrade the return loss if not carefully controlled.

7.2 Component Tolerance Sensitivity

The performance of a resonant filter is highly sensitive to component variation. If the inductors or capacitors deviate from their nominal values, the center frequency will shift and the matching condition may weaken. For a resonance defined by

$$f_0 = \frac{1}{2\pi\sqrt{LC}}, \tag{7.1}$$

a small perturbation in either L or C produces a corresponding shift in the resonant frequency. This sensitivity becomes more important as the filter order increases or as the operating band moves higher in frequency.

7.3 Practical Applications

Reflectionless band pass filters are useful in receiver front ends, test equipment, spectrum-monitoring systems, and cascaded RF chains where the input match is as important as the spectral response. Their broadband absorptive behavior makes them particularly attractive in systems that are vulnerable to reverse reflections or that contain multiple passive and active blocks in series.

7.4 Interpretation of the Topology

The main conceptual lesson of the reflectionless approach is that a filter need not choose between selectivity and matching. By embedding the correct resistive paths into a symmetric network, the rejected energy can be controlled internally. This perspective changes the role of the filter from a purely frequency-selective element to an interface-control element in the overall RF system.

Chapter 8

Conclusion and Future Work

8.1 Conclusion

This thesis presented the design and analysis of absorptive reflectionless band pass filters using a structured synthesis approach grounded in even-odd mode analysis, symmetry, ladder duality, and frequency scaling. The work showed how a single-stage reflectionless cell can serve as the basis for more complex cascaded and scaled implementations. Across all variants, the same core advantage was observed: unwanted energy is absorbed rather than reflected, which improves broadband matching and reduces inter-stage interaction.

The study also confirmed that reflectionless filtering is not a niche theoretical idea but a practical RF design strategy. It offers a useful way to combine selectivity with impedance control, making it well suited to modern cascaded RF systems. The comparative behavior of the single-stage and cascaded versions demonstrates that the topology can be adapted to different design goals without abandoning its absorptive character.

8.2 Future Work

Several extensions can be pursued in future work. First, the current lumped-element framework can be developed into a full PCB implementation with measured hardware data. Second, the sensitivity of the topology to tolerance spread and parasitic effects can be studied in greater detail. Third, the same reflectionless methodology can be extended to higher-order or wider-bandwidth responses. Finally, active or hybrid compensation techniques may be investigated to reduce insertion loss while preserving the absorptive behavior.

Appendix A

Appendix A: Formula Summary

This appendix collects the principal formulas used throughout the report. They are grouped here for quick reference during synthesis, scaling, and response interpretation.

A.1 Two-Port Network Relations

The S-parameters of a two-port network are defined by

$$S_{11} = \left.\frac{b_1}{a_1}\right|_{a_2=0}, \qquad S_{21} = \left.\frac{b_2}{a_1}\right|_{a_2=0}, \tag{A.1}$$

$$S_{12} = \left.\frac{b_1}{a_2}\right|_{a_1=0}, \qquad S_{22} = \left.\frac{b_2}{a_2}\right|_{a_1=0}. \tag{A.2}$$

The input reflection coefficient and return loss are

$$\Gamma = \frac{Z_{\text{in}} - Z_0}{Z_{\text{in}} + Z_0}, \qquad RL(f) = -20\log_{10}|S_{11}(f)|\,. \tag{A.3}$$

A.2 Energy and Symmetry Relations

The passive power balance and even-odd mode decomposition are

$$P_{\text{in}} = P_{\text{load}} + P_{\text{loss}} + P_{\text{ref}}, \qquad V_e = \frac{V_1 + V_2}{2}, \quad V_o = \frac{V_1 - V_2}{2}, \tag{A.4}$$

$$I_e = \frac{I_1 + I_2}{2}, \qquad I_o = \frac{I_1 - I_2}{2}. \tag{A.5}$$

A.3 Band-Pass Synthesis Relations

For a band-pass filter with cut-off frequencies f_{c1} and f_{c2},

$$f_0 = \sqrt{f_{c1} f_{c2}}, \qquad BW = f_{c2} - f_{c1}, \qquad FBW = \frac{BW}{f_0}. \tag{A.6}$$

The transmission-zero span is

$$BW_{tz} = f_{tz2} - f_{tz1}. \tag{A.7}$$

The low-pass to band-pass transformation is

$$s \rightarrow \frac{s^2 + \omega_0^2}{Bs}. \tag{A.8}$$

For a normalized prototype, the resonant element relations are

$$L_s = \frac{R_0 g_k}{\omega_0 FBW}, \qquad C_s = \frac{FBW}{R_0 \omega_0 g_k}, \tag{A.9}$$

$$L_p = \frac{R_0 FBW}{\omega_0 g_k}, \qquad C_p = \frac{g_k}{R_0 \omega_0 FBW}. \tag{A.10}$$

A.4 Frequency Scaling and Cascade Relations

For frequency and impedance scaling factors K_f and K_z,

$$L' = \frac{K_z}{K_f} L, \qquad C' = \frac{1}{K_z K_f} C. \tag{A.11}$$

For a cascade of N stages,

$$H_N(s) = \prod_{k=1}^{N} H_k(s), \qquad \text{and for identical stages, } H_N(s) = \left[H_1(s)\right]^N. \tag{A.12}$$

The resonant frequency sensitivity to L and C is

$$f_0 = \frac{1}{2\pi\sqrt{LC}}. \tag{A.13}$$

A.5 ABCD Representation

The ABCD matrix of a two-port is defined as

$$\begin{pmatrix} V_1 \\ I_1 \end{pmatrix} = \begin{pmatrix} A & B \\ C & D \end{pmatrix} \begin{pmatrix} V_2 \\ -I_2 \end{pmatrix}. \tag{A.14}$$

This representation is useful for cascading multiple sections because the overall matrix is obtained by matrix multiplication.

Appendix B

Appendix B: Prototype Coefficients

The following normalized low-pass prototype coefficients are included because they provide the starting point for the band-pass synthesis relations used in Appendix A. The coefficients are not arbitrary constants; they represent the ladder prototype values that determine the resonant element ratios after frequency transformation and impedance scaling. In practical design, the selected order and prototype family influence the resulting selectivity, passband flatness, and the location of transmission zeros.

For a reflectionless band-pass implementation, these coefficients are combined with the target cut-off frequencies and reference impedance to determine the inductor and capacitor values for the resonant sections. The coefficients below are representative values used for low-order prototype synthesis and are sufficient for illustrating the design flow presented in this thesis.

Table B.1: Illustrative prototype coefficients.

Order	g_1	g_2	g_3
1	1.0	-	-
2	1.1468	1.0	-
3	1.3968	1.8015	1.1468
4	1.5963	1.0967	1.5963

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